

EU Code - EU Title

Cycle: **Engineer**

Option/Sector: **Computer Science and Information Systems**

Constituent Element (CE) Code: **ING4178**

EC title: **ISI Transversal project**

Level: **4**

Semester: **2**

Credits: **4**

Language of instruction: **EN**

CM: **10h**

TD: **0h**

PW: **0h**

FTF: **30h**

TrEN : **20h**

Autonomy: **goh**

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GENERAL PRESENTATION OF THE COURSE

Course objectives:

The objective of the EU is to understand and apply all the concepts and skills acquired during the academic year. The practice of these skills will be carried out on a project that extends over 02 semesters. In the first semester, the student carried out a project using the methods and techniques of software engineering, architecture of an Information System and advanced concepts in web architecture. While in the second semester, it will be a question of improving these techniques through new skills that he will have to acquire. He will therefore have to implement a more solid and well-structured digital platform in order to ensure at least the quality of the deliverables that he will deliver; it will also be a question of further refining these skills in terms of rendering software.

Prerequisites:

Software engineering, IS Architecture, Web architecture and programming 1, OOP, basic security, web services (APIs).

SKILLS TARGETED

At the end of this course, based on a specification of needs or specifications, the student should be able to:

- Implement a solution that meets user needs;
- Ensure that the developed platform meets a solid and layered architecture;
- Ensure the platform is deployed
- Implement basic security;
- Teamwork using a collaborative approach through agility;
- Carry out technologies and concepts (AI, Generative AI, ML, LLM etc.) ;

COURSE ORGANIZATION

Overall course content

- Follow-up session on the progress of the project
- Support and improvement of work produced
- Students will present their work every 2 weeks

ASSESSMENT

	Written or Practical	Individual or Group	Percentage of overall rating
Follow up	Practical	Individual and Group	40%
Presentation (FP)	Practical	Group 04 or 05	60%

COURSE METHODOLOGY

- Students will present the progress of their work in oral and visual form. The presentation of the work leads to an evaluation of the students.

TEACHING MATERIALS AND WORKING ENVIRONMENT

- Laptop capable of handling the implementation load with .pdf and .ppt file reader software installed.
- Paper support and tools needed for note-taking and calculations for activities.
- Spring Boot Framework
- Angular 18 Framework
- React Native Framework
- Node.js Framework

RECOMMENDED BIBLIOGRAPHY

- Mobile App Development with Ionic: Create Robust and Intuitive Cross-Platform Mobile Apps - Charles Edou NZE.
- Mastering Angular-15 and Ionic-7: A Comprehensive Guide to Building Your First Powerful Mobile App (E-book) – Karan Gupta, 2023
- Mastering Ionic: A Comprehensive Guide to Learn Ionic, by Cybellium Ltd and Kris Hermans
- React Native for Mobile Development - Akshat Paul & Abhishek Nalwaya , 2019. Appress .
- React and React Native: Build cross-platform JavaScript and TypeScript apps for the web, desktop, and mobile - Adam Boduch , Roy Derks & Mikhail Sakhniuk , 2024.Packt.